FLUXOR — Final Benchmark Summary

27 benchmark trials: 3 algorithms × 3 scenarios × 3 trials. Values are means across the three trials.

Algorithm	Scenario	Throughput	Avg ms	P95 ms	Error %	CPU STDDEV %
RL	Normal	314.44	58.39	81.51	0.00	0.90
RL	Spike	891.25	57.39	81.60	0.00	0.90
RL	Failure	313.41	58.96	83.42	0.34	3.53
RR	Normal	313.73	58.70	81.97	0.00	1.03
RR	Spike	884.58	58.58	83.09	0.00	0.81
RR	Failure	312.25	59.43	83.74	0.20	3.29
LC	Normal	313.27	58.95	82.95	0.00	0.80
LC	Spike	883.52	58.77	82.76	0.00	0.75
LC	Failure	312.22	59.53	83.24	0.27	3.20

Key observations

• RL achieved the highest throughput in all three measured scenarios.

• Under spike load, RL achieved 891.25 req/s, 57.39 ms average latency, and 81.60 ms P95 latency.

• Under failure, RL achieved 313.41 req/s and 58.96 ms average latency.

• LC had lower CPU utilization STDDEV than RL in all three scenarios.

• Projected performance targets should be replaced by these measured experimental results in the paper.

RL vs baselines — spike scenario

Throughput: +0.75% vs RR and +0.87% vs LC. Average latency: 2.03% lower vs RR and 2.35% lower vs LC. P95 latency: 1.80% lower vs RR and 1.41% lower vs LC.